

Network Troubleshooting

Final Lab Report

Kyle School Project Portfolio

Item	Details
Project Folder	Networking / Network-Troubleshooting
Primary Focus	Connectivity, routing, DNS, SSH, Proxmox access, Packet Tracer, and service troubleshooting
Platforms	Proxmox, Cisco Packet Tracer, Windows, Ubuntu Server, Kali Linux, Windows Server
Common Lab Networks	10.10.10.0/24, 10.10.10.0/16, 192.168.100.0/24, 192.168.200.0/24
Report Purpose	Document troubleshooting methodology, tools, issues, resolutions, and professional project outcomes

Disclaimer

This report documents authorized troubleshooting and testing performed in educational lab environments. The processes described are intended for systems owned, assigned, or controlled within the lab.

1. Executive Summary

This project documents a structured network troubleshooting workflow developed from the Kyle School networking, server, virtualization, and cybersecurity labs. The work involved diagnosing and resolving connectivity issues across Windows systems, Linux systems, Proxmox infrastructure, Cisco Packet Tracer environments, web services, SSH services, DNS, routing, and virtual networks.

The project demonstrates the ability to identify symptoms, gather evidence, isolate root causes, apply corrective actions, and validate successful resolution. Troubleshooting activities included IP configuration review, gateway testing, DNS validation, routing checks, Proxmox bridge verification, Cisco VLAN and trunk troubleshooting, service status review, firewall validation, and packet capture analysis.

2. Lab Environment

Component	Details
Project Folder	Networking / Network-Troubleshooting
Virtualization	Proxmox Virtual Environment
Network Simulation	Cisco Packet Tracer
Operating Systems	Windows 10/11, Ubuntu Server, Kali Linux, Windows Server, Proxmox VE
Cisco Devices	Cisco 2911 routers, Cisco 2960 switches, Packet Tracer PCs and wireless devices
Example Proxmox Host	team2.tech - 10.10.10.88/24
Example Windows Client	10.10.10.55/24
Example DVWA Server	UbunServDVWA-TeamA - 10.10.10.223/16

3. Project Objectives

- Create a repeatable network troubleshooting workflow.
- Validate local and remote connectivity using command-line tools.
- Diagnose IP address, subnet, gateway, DNS, and routing problems.
- Troubleshoot Proxmox GUI and SSH connectivity issues.
- Troubleshoot Linux SSH and service availability problems.
- Troubleshoot Cisco Packet Tracer VLAN, trunk, router, subinterface, and routing issues.
- Use packet capture tools to validate traffic flow.
- Document issue, cause, resolution, and validation steps in a professional format.

4. Troubleshooting Methodology

Step	Purpose	Example Tools
1. Identify the symptom	Define what is not working and what error is observed.	Browser error, ping result, SSH error, Packet Tracer message
2. Check local configuration	Verify local IP, subnet, interface, and adapter state.	ip a, ipconfig /all, show ip interface brief
3. Test gateway	Determine whether the local host can reach its network point of exit.	ping <gateway>, ip route, route print
4. Test destination by IP	Separate routing/connectivity issues from DNS issues.	ping <target-ip>, tracert, traceroute
5. Test DNS	Confirm hostname resolution.	nslookup, resolvectl status, /etc/resolv.conf

6. Check service and port	Confirm the required service is running and listening.	systemctl, ss -tulnp, lsof -i
7. Review firewall/rules	Confirm traffic is not blocked by firewall or ACL.	ufw status, iptables, show access-lists
8. Capture traffic	Validate whether packets are reaching the expected interface.	tcpdump, Wireshark
9. Apply fix and validate	Make one change at a time and retest.	ping, browser access, SSH login, Packet Tracer simulation

5. Work Performed

5.1 Physical and Virtual Connectivity Checks

- Verified active network adapters, correct virtual switch or bridge assignment, and working links.
- Reviewed Proxmox bridge behavior, Windows adapter configuration, Linux interface status, and Packet Tracer cabling/link lights.

5.2 IP Addressing Validation

- Reviewed IP addresses, subnet masks, default gateways, and DNS settings.
- Used ip a, ip route, ipconfig /all, ping, and Cisco show commands to validate configuration.

5.3 Connectivity Testing

- Tested loopback, gateway, same-subnet, cross-subnet, server, and web interface connectivity.
- Used ping results to identify whether failures were local, gateway-related, routing-related, or service-related.

5.4 DNS Troubleshooting

- Investigated cases where systems were reachable by IP address but not hostname or domain name.
- Reviewed DNS client settings, resolver output, nslookup results, and local name behavior.

5.5 Routing and Gateway Troubleshooting

- Reviewed routing tables and default gateway configuration across Windows, Linux, Proxmox, and Cisco labs.
- Used route print, ip route, tracert, traceroute, show ip route, and Packet Tracer simulation tests.

5.6 Proxmox Access Troubleshooting

- Troubleshoot loss of browser and SSH access to the Proxmox server.
- Validated the Proxmox IP address, Windows static IP address, subnet alignment, ping results, DNS behavior, and management URL.

5.7 Linux SSH and Service Troubleshooting

- Reviewed SSH service status, listening ports, firewall rules, and authentication logs.
- Used systemctl, journalctl, ss, and lsof to confirm whether the service was active and reachable.

5.8 Packet Tracer VLAN and Router Troubleshooting

- Resolved VLAN, trunk, router-on-a-stick, DHCP, ACL, and inter-VLAN routing problems in Packet Tracer.
- Validated Cisco 2911 router and Cisco 2960 switch configurations using show commands and ping tests.

5.9 Packet Capture and Traffic Review

- Used tcpdump and Wireshark to verify traffic flow and support investigation.
- Captured SSH and HTTP traffic during authorized lab testing when needed.

6. Tools and Commands Used

Category	Tools / Commands
Windows	ipconfig /all, ping, tracert, nslookup, route print
Linux	ip a, ip route, ping, resolvectl status, systemctl, journalctl, ss -tulnp, lsof -i
Cisco	show vlan brief, show interfaces trunk, show ip interface brief, show running-config, show ip route, show access-lists
Packet Capture	tcpdump, Wireshark
Proxmox	Web GUI, Linux bridge review, host IP verification
Service Testing	curl, browser testing, SSH client testing

7. Issues Encountered and Resolutions

Issue	Likely Cause	Resolution / Validation
Destination host unreachable	Incorrect IP, subnet, gateway, bridge, or route	Reviewed IP settings, gateway, bridge assignment, and ping path.
Proxmox web GUI not reachable	Client not in correct subnet, DNS issue, or wrong URL	Set correct IP, client workstation in correct range, pinged Proxmox, verified browser URL
DNS resolution failure	DNS server or resolver misconfiguration	Tested with nslookup, reviewed resolver configuration, used IP access while DNS was resolved
SSH unavailable	SSH service stopped, firewall block, wrong IP, or authentication issue	Checked if service was running, ss -tulnp, firewall rules, and auth logs.
Packet Tracer subinterface error	IP applied before VLAN encapsulation	Configured subinterface encapsulation first, then applied IP address and tested interface
Web service unreachable	Apache stopped, wrong IP, firewall block, or wrong service status	Checked service status, listening ports, curl/browser response, and firewall rules

8. Final Outcome

The project produced a repeatable troubleshooting workflow that can be used across IT support, networking, server administration, cybersecurity, and virtualization environments. It demonstrates practical troubleshooting skills across multiple platforms instead of a single isolated tool or device.

- Connectivity problems were isolated using local, gateway, destination, DNS, and service checks.
- Proxmox access issues were resolved through IP addressing and client configuration validation.
- Linux service issues were reviewed using service, port, and log commands.
- Cisco Packet Tracer network issues were resolved using VLAN, trunk, subinterface, route, and ACL verification.
- Packet capture tools were used to support validation and investigation.
- Documentation was created for GitHub portfolio and resume project use.

9. Recommended Evidence

- IP configuration screenshots from Windows and Linux.
- Ping test results showing before/after troubleshooting.
- Routing table outputs.

- DNS lookup outputs.
- SSH or web service status outputs.
- Proxmox GUI access screenshot.
- Packet Tracer VLAN and trunk verification screenshots.
- Wireshark or tcpdump packet capture screenshots.

10. Recommended GitHub Folder Structure

Networking/Network-Troubleshooting/

README.md

final-lab-report.pdf

commands.txt

notes.md

evidence/

ip-configuration.txt

routing-table.txt

ping-results.txt

dns-results.txt

service-status.txt

screenshots/

ip-configuration.png

ping-test.png

routing-table.png

proxmox-gui-access.png

packet-tracer-verification.png

11. Reflection

This project reinforced the value of a structured troubleshooting process. Instead of changing random settings, the work showed the importance of testing one layer at a time: local interface, IP settings, gateway, destination, DNS, route, firewall, service, and packet flow. The project also emphasized that strong documentation is part of professional troubleshooting. Recording commands, outputs, symptoms, fixes, and validation steps makes the work easier to explain in a classroom, job interview, resume, or GitHub portfolio.